

Maths - Question 75

Original paper page 62; original question and option regions

$$\int_{-3}^3 |2 - x| \, dx =$$

Maths - Question 75 (continued)

Original paper page 63; original question and option regions

ons :



12

16

13

25



Maths - Question 76

Original paper page 63; original question and option regions

$$\int_{\frac{1}{\sqrt[5]{31}}}^{\frac{1}{\sqrt[5]{242}}} \frac{1}{\sqrt[5]{x^{30} + x^{25}}} dx =$$

Options :

$$\frac{65}{4}$$

1. ✖

$$\frac{-75}{4}$$

2. ✖

Maths - Question 76 (continued)

Original paper page 64; original question and option regions

$$\frac{75}{4}$$

$$\frac{-65}{4}$$



Maths - Question 77

Original paper page 64; original question and option regions

Area of the region (in square units) enclosed by the curves $y^2 = 8(x + 2)$, $y^2 = 4(1 - x)$ and the Y-axis is

$y^2 = 8(x + 2)$, $y^2 = 4(1 - x)$ వక్రములు మరియు Y-అక్షములచే ఆవరించబడిన ప్రాంతం యొక్క వైశాల్యం (చ. యూనిట్లలో)

Options :

$$\frac{8}{3}(5 - 3\sqrt{2})$$

1. ✓

$$\frac{8}{3}(\sqrt{2} - 1)$$

2. ✗

$$\frac{8}{3}(3 - \sqrt{2})$$

3. ✗

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Maths - Question 77 (continued)

Original paper page 65; original question and option regions

$$\frac{4}{3}(\sqrt{2} + 1)$$



Maths - Question 78

Original paper page 65; original question and option regions

The sum of the order and degree of the differential equation $\frac{d^4y}{dx^4} = \left\{ c + \left(\frac{dy}{dx} \right)^2 \right\}^{3/2}$ is

అవకలన సమీకరణము $\frac{d^4y}{dx^4} = \left\{ c + \left(\frac{dy}{dx} \right)^2 \right\}^{3/2}$ యొక్క పరిమాణము మరియు తరగతుల మొత్తము

Options :

1. ✖ 4

2. ✔ 6

3. ✖ 5

4. ✖ 8

The general solution of the differential equation $(x + y)y \, dx + (y - x)x \, dy = 0$ is

$(x + y)y \, dx + (y - x)x \, dy = 0$ అవకలన సమీకరణం యొక్క సాధారణ సాధన

Options :

$$x + y \log(cy) = 0$$

1. ✖

$$\frac{y}{x} = \log(xy) + c$$

2. ✖

$$x + y \log(cxy) = 0$$

3. ✔

$$\frac{y}{x} = \log(cxy)$$

4. ✖

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Maths - Question 80

Original paper page 66; original question and option regions

The general solution of the differential equation $(y^2 + x + 1)dy = (y + 1)dx$ is

$(y^2 + x + 1)dy = (y + 1)dx$ అవకలనీయ సమీకరణం యొక్క సాధారణ సాధన

Options :

$$x + 2 + (y + 1) \log(y + 1)^2 = y + c$$

1. ✖

Maths - Question 80 (continued)

Original paper page 67; original question and option regions



$$x + 2 + \log(y + 1)^2 = \frac{y}{y + 1} + c$$

$$\frac{x}{y + 1} = \log(y + 1)^2 + y + c$$

$$\frac{x + 2}{y + 1} + \log(y + 1)^2 = y + c$$

Maths - Question 1

Original paper page 2; original question and option regions

$f: \mathbb{R} \rightarrow \mathbb{R}$ is defined by $f(x+y) = f(x) + 12y, \forall x, y \in \mathbb{R}$. If $f(1) = 6$, then

$$\sum_{r=1}^n f(r) =$$

$f: \mathbb{R} \rightarrow \mathbb{R}$ అనేది $f(x+y) = f(x) + 12y, \forall x, y \in \mathbb{R}$ గా నిర్వచించబడింది. $f(1) = 6$ అయితే,

$$\sum_{r=1}^n f(r) =$$

Options :

1. ✖ n^2

Maths - Question 1 (continued)

Original paper page 3; original question and option regions



$$5n^2$$

$$6n^2$$

$$\frac{3n(n+1)}{2}$$

Maths - Question 2

Original paper page 3; original question and option regions

The domain of the real valued function $f(x) = \sqrt{2+x} + \sqrt{3-x}$ is

$f(x) = \sqrt{2+x} + \sqrt{3-x}$ వాస్తవ మూల్య ప్రమేయం యొక్క ప్రదేశం

Options :

1. ✖ $(-2, 3)$

2. ✖ $[-2, 3)$

3. ✖ $(-2, 3]$

4. ✔ $[-2, 3]$

Question Number : 3 Question Id : 4509386723 Display Question Number : Yes Is Question

Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Inst
Time : 0

If $2 \cdot 4^{2n+1} + 3^{3n+1}$ is divisible by k for all $n \in N$, then $k =$

ప్రతి $n \in N$ కు, $2 \cdot 4^{2n+1} + 3^{3n+1}$ అనేది k చే భాగింపబడితే, $k =$

Options :

1. ✖ 209

2. ✔ 11

3. ✖ 8

4. ✖ 3

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Maths - Question 4

Original paper page 4; original question and option regions

$$\begin{vmatrix} a & b & c \\ a^2 & b^2 & c^2 \\ 1 & 1 & 1 \end{vmatrix} \text{ is not equal to}$$

$$\begin{vmatrix} a & b & c \\ a^2 & b^2 & c^2 \\ 1 & 1 & 1 \end{vmatrix} \text{ కు సమానము కానిది}$$

Options :

$$\begin{vmatrix} a+1 & b+1 & c+1 \\ a^2+1 & b^2+1 & c^2+1 \\ 1 & 1 & 1 \end{vmatrix}$$

1. ✖

Maths - Question 4 (continued)

Original paper page 5; original question and option regions



$$\begin{vmatrix} a-b & b-c & c \\ a^2-b^2 & b^2-c^2 & c^2 \\ 0 & 0 & 1 \end{vmatrix}$$

$$\begin{vmatrix} a(a+1) & b(b+1) & c(c+1) \\ a+1 & b+1 & c+1 \\ -1 & -1 & -1 \end{vmatrix}$$

$$\begin{vmatrix} a+b & b+c & c+a \\ a^2+b^2 & b^2+c^2 & c^2+a^2 \\ 2 & 2 & 2 \end{vmatrix}$$

Maths - Question 5

Original paper page 5; original question and option regions

Let A, B, C, D and E be $n \times n$ matrices, each with non – zero determinant. If $ABCDE = I$,
then $C^{-1} =$

A, B, C, D మరియు E లు $n \times n$ చతురస్ర మాత్రికలు, ప్రతి మాత్రిక శూన్యేతర నిర్ధారకం కలిగిన మాత్రిక. $ABCDE = I$, అయితే $C^{-1} =$

Options :

$$E^{-1}D^{-1}B^{-1}A^{-1}$$

1. ✖

$$DEAB$$

2. ✔

Maths - Question 5 (continued)

Original paper page 6; original question and option regions

$$A^{-1}B^{-1}D^{-1}E^{-1}$$



ABDE

4. ✖

Maths - Question 6

Original paper page 6; original question and option regions

If $A = [a_{ij}]$, $1 \leq i, j \leq n$ with $n \geq 2$ and $a_{ij} = i + j$ is a matrix, then the rank of A is

$n \geq 2$ మరియు $a_{ij} = i + j$ అయ్యేటట్లు $A = [a_{ij}]$, $1 \leq i, j \leq n$ అనేది ఒక మాత్రిక అయితే మాత్రిక A యొక్క ర్యాంక్

Options :

1. ✖ 0

2. ✖ 1

3. ✔ 2

4. ✖ 4

Maths - Question 7

Original paper page 7; original question and option regions



If $z_1 = 10 + 6i$, $z_2 = 4 + 6i$ and z is any complex number such that the argument of $\frac{(z-z_1)}{(z-z_2)}$ is $\frac{\pi}{4}$,

then

$z_1 = 10 + 6i$, $z_2 = 4 + 6i$ మరియు $\frac{(z-z_1)}{(z-z_2)}$ యొక్క ఆయామం $\frac{\pi}{4}$ అయ్యేటట్లుగా z ఏదైనా

సంకీర్ణ సంఖ్య అయితే, అప్పుడు

Options :

$$|z - 7 - 9i| = 3\sqrt{2}$$

1. ✓

$$|z - 7 - 9i| = 2\sqrt{2}$$

2. ✗

$$|z - 3 + 9i| = 3\sqrt{2}$$

3. ✗

$$|z + 3 - 9i| = 2\sqrt{2}$$

4. ✗

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Maths - Question 8

Original paper page 7; original question and option regions

If $\frac{3-2i \sin \theta}{1+2i \sin \theta}$ is purely imaginary number, then $\theta =$

$\frac{3-2i \sin \theta}{1+2i \sin \theta}$ పుద్ధకల్పిత సంఖ్య అయితే $\theta =$

Options :

$$2n\pi \pm \frac{\pi}{4}$$

1. ✖

Maths - Question 8 (continued)

Original paper page 8; original question and option regions



$$2n\pi \pm \frac{\pi}{2}$$

$$n\pi \pm \frac{\pi}{3}$$

$$n\pi \pm \frac{\pi}{6}$$

Maths - Question 9

Original paper page 8; original question and option regions

If $z = x+iy$, $x^2+y^2 = 1$ and $z_1 = ze^{i\theta}$, then $\frac{z_1^{2n}-1}{z_1^{2n}+1} =$

$z = x+iy$, $x^2+y^2=1$ మరియు $z_1=ze^{i\theta}$ అయిన $\frac{z_1^{2n}-1}{z_1^{2n}+1} =$

Options :

$$-i \tan \left(n \left(\theta + \tan^{-1} \left(\frac{y}{x} \right) \right) \right)$$

1. ✖

$$i \cot \left(n \left(\theta + \tan^{-1} \frac{y}{x} \right) \right)$$

2. ✖

Maths - Question 9 (continued)

Original paper page 9; original question and option regions



$$i \tan \left(n \left(\theta + \tan^{-1} \frac{x}{y} \right) \right)$$

$$i \tan \left(n \left(\theta + \tan^{-1} \frac{y}{x} \right) \right)$$

4. ✓

Maths - Question 10

Original paper page 9; original question and option regions

Let $[r]$ denote the largest integer not exceeding r and the roots of the equation

$3x^2 + 6x + 5 + \alpha(x^2 + 2x + 2) = 0$ are complex numbers whenever $\alpha > L$ and $\alpha < M$. If

$(L-M)$ is minimum, then greatest value of $[r]$ such that $Ly^2 + My + r < 0$ for all $y \in \mathbb{R}$ is

r కి మించని గరిష్ఠ పూర్ణాంకాన్ని $[r]$ సూచిస్తుందని, $\alpha > L$ మరియు $\alpha < M$ అయినప్పుడు,

$3x^2 + 6x + 5 + \alpha(x^2 + 2x + 2) = 0$ సమీకరణం యొక్క మూలాలు సంకీర్ణ సంఖ్యలు అని

అనుకోండి. $(L-M)$ కనిష్ఠం అయితే, ప్రతి $y \in \mathbb{R}$ కి, $Ly^2 + My + r < 0$ అయ్యేటట్లుగా ఉండే $[r]$

యొక్క గరిష్ఠ విలువ

Options :

1. ✓ L

2. ✗ M

3. ✗ $L + M$

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Maths - Question 10 (continued)

Original paper page 10; original question and option regions

$$M - L$$



Maths - Question 11

Original paper page 10; original question and option regions

For any real value of x , if $\frac{11x^2+12x+6}{x^2+4x+2} \notin (a, b]$ then the value x for which

$$\frac{11x^2+12x+6}{x^2+4x+2} = b - a + 3 \text{ is}$$

ఏ వాస్తవ సంఖ్య x కైనా $\frac{11x^2+12x+6}{x^2+4x+2} \notin (a, b]$ అయితే, $\frac{11x^2+12x+6}{x^2+4x+2} = b - a + 3$ అయ్యేటట్లు ఉన్న x విలువ

Options :

1. ✖ $\frac{3}{4}$

2. ✖ $\frac{3}{2}$

3. ✖ 2

4. ✔ $-\frac{1}{2}$

Time : 0



If the roots of $\sqrt{\frac{1-y}{y}} + \sqrt{\frac{y}{1-y}} = \frac{5}{2}$ are α and β ($\beta > \alpha$) and the equation

$(\alpha + \beta)x^4 - 25\alpha\beta x^2 + (\gamma + \beta - \alpha) = 0$ has real roots, then a possible value of γ is

$\sqrt{\frac{1-y}{y}} + \sqrt{\frac{y}{1-y}} = \frac{5}{2}$ యొక్క మూలాలు α మరియు β ($\beta > \alpha$) మరియు సమీకరణం

$(\alpha + \beta)x^4 - 25\alpha\beta x^2 + (\gamma + \beta - \alpha) = 0$ కు వాస్తవ మూలాలు ఉంటే, γ కు సాధ్యమయ్యే ఒక

విలువ

Options :

1. ✓ $\frac{1}{2}$

2. ✗ 4

3. ✗ 2π

4. ✗ $\sqrt{e + 13}$

Maths - Question 13

Original paper page 11; original question and option regions

If the roots of the equation $x^3 + ax^2 + bx + c = 0$ are in arithmetic progression, then

సమీకరణము $x^3 + ax^2 + bx + c = 0$ యొక్క మూలాలు అంకశ్రేణి లో ఉంటే, అప్పుడు

Maths - Question 13 (continued)

Original paper page 12; original question and option regions



$$a^3 - 3ab + c = 0$$

$$9ab = 2a^3 + 27c$$



$$a^2 - 2bc + c = 0$$



$$3ab - 3c - a^3 = 0$$



Maths - Question 14

Original paper page 12; original question and option regions

A test containing 3 objective type of questions is conducted in a class. Each question has 4 options and only one option is the correct answer. No two students of the class have answered identically and no student has written all correct answers. If every student has attempted all the questions, then the maximum possible number of students who has written the test is

ఒక తరగతిలో 3 ఐచ్చిక తరహా ప్రశ్నలు గల ఒక టెస్ట్ ను నిర్వహించారు. ప్రతి ప్రశ్నకు 4 ఐచ్చికాలుంటాయి మరియు ఒకే ఐచ్చికం సరైన సమాధానంగా ఉంటుంది. ఏ ఇద్దరు విద్యార్థుల సమాధానాలూ సర్వసమానంగా లేవు మరియు ఏవిద్యార్థి అన్ని సరియైన సమాధానాలు వ్రాయలేదు. ప్రతి విద్యార్థి అన్ని ప్రశ్నలను ప్రయత్నిస్తే, ఆ టెస్ట్ ను వ్రాయడానికి సాధ్యమయ్యే విద్యార్థుల గరిష్ఠ సంఖ్య

Options :

80

1. ✖

Maths - Question 14 (continued)

Original paper page 13; original question and option regions

63

15

11



Maths - Question 15

Original paper page 13; original question and option regions

The number of numbers lying between 1000 and 10000 such that every number contains the digits 3 and 7 only once without repetition is

1000 మరియు 10000 మధ్య ఉంటూ, ప్రతి సంఖ్యలోను 3 మరియు 7 అంకెలను పునరావృతం కాకుండా ఒక్కసారి మాత్రమే కలిగి ఉండే సంఖ్యల సంఖ్య

Options :

1. ✖ 1140

2. ✖ 918

3. ✔ 720

4. ✖ 810

Mandatory : No Calculator : None Response Time : N.A Think Time : N.A**Time : 0**

The number of ways in which 17 apples can be distributed among four guests such that each guest gets at least 3 apples is

17 ఆపిల్ పండ్లను నలుగురు అతిథులకు ప్రతి అతిథి కనీసం 3 ఆపిల్ పండ్లను పొందేటట్లుగా పంచగలిగే విధాలసంఖ్య

Options :

1. ✖ 1140

2. ✖ 336

3. ✖ 36

4. ✔ 56

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Maths - Question 17

Original paper page 14; original question and option regions

If the coefficients of x^5 and x^6 are equal in the expansion of $(a + \frac{x}{5})^{65}$, then the coefficient of x^2 in the expansion of $(a + \frac{x}{5})^4$ is

$(a + \frac{x}{5})^{65}$ విస్తరణ లో x^5 మరియు x^6 ల యొక్క గుణకాలు సమానమైతే అప్పుడు $(a + \frac{x}{5})^4$ యొక్క విస్తరణ లో x^2 యొక్క గుణకం

Maths - Question 17 (continued)

Original paper page 15; original question and option regions

1

$$\frac{32}{25}$$

2

$$\frac{24}{25}$$

Maths - Question 18

Original paper page 15; original question and option regions

If $|x| < \frac{2}{3}$, then the 4th term in the expansion of $(3x - 2)^{2/3}$ is

$|x| < \frac{2}{3}$ అయితే, $(3x - 2)^{2/3}$ విస్తరణ లోని 4 వ పదం

Options :

$$\frac{\sqrt[3]{4}}{6} x^3$$

1. ✖

$$-\frac{\sqrt[3]{4}}{6} x^3$$

2. ✔

Maths - Question 18 (continued)

Original paper page 16; original question and option regions



$$\frac{\sqrt[3]{4}}{8}x^3$$

$$-\frac{\sqrt[3]{4}}{8}x^3$$

4. ✖

Maths - Question 19

Original paper page 16; original question and option regions

If $\frac{x^2+3}{x^4+2x^2+9} = \frac{Ax+B}{x^2+ax+b} + \frac{Cx+D}{x^2+cx+b}$ then $aA + bB + cC + D =$

$\frac{x^2+3}{x^4+2x^2+9} = \frac{Ax+B}{x^2+ax+b} + \frac{Cx+D}{x^2+cx+b}$ అయితే, $aA + bB + cC + D =$

Options :

1. ✖ 1

2. ✖ 0

3. ✖ -1

4. ✔ 2

Mandatory : No Calculator : None Response Time : N.A Think Time : N.A**Time : 0**

If $\sec \theta + \tan \theta = \frac{1}{3}$, then the quadrant in which 2θ lies is

$\sec \theta + \tan \theta = \frac{1}{3}$ అయితే, 2θ ఉండే పాదం

Options :1st quadrant

1 వ పాదం

1. ✖

2nd quadrant

2 వ పాదం

2. ✖

3rd quadrant

3 వ పాదం

3. ✔

4th quadrant

4 వ పాదం

4. ✖

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Maths - Question 21

Original paper page 17; original question and option regions

If $540^\circ < A < 630^\circ$ and $|\cos A| = \frac{5}{13}$ then $\tan \frac{A}{2} \tan A =$

$540^\circ < A < 630^\circ$ మరియు $|\cos A| = \frac{5}{13}$ అయితే, $\tan \frac{A}{2} \tan A =$

Maths - Question 21 (continued)

Original paper page 18; original question and option regions

$$\frac{18}{5}$$

$$\frac{8}{5}$$

$$-\frac{8}{5}$$

$$-\frac{18}{5}$$

Maths - Question 22

Original paper page 18; original question and option regions

If $(\alpha + \beta)$ is not a multiple of $\frac{\pi}{2}$ and $3 \sin(\alpha - \beta) = 5 \cos(\alpha + \beta)$, then

$$\tan\left(\frac{\pi}{4} + \alpha\right) + 4 \tan\left(\frac{\pi}{4} + \beta\right) =$$

$(\alpha + \beta)$ అనేది $\frac{\pi}{2}$ యొక్క గుణిజం కాదు మరియు $3 \sin(\alpha - \beta) = 5 \cos(\alpha + \beta)$ అయితే,

$$\tan\left(\frac{\pi}{4} + \alpha\right) + 4 \tan\left(\frac{\pi}{4} + \beta\right) =$$

Options :

1. ✓

0

2. ✗

1

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Maths - Question 22 (continued)

Original paper page 19; original question and option regions

4



2



The general solution of the equation $\sin^2\theta + 3\cos^2\theta = 5\sin\theta$ is

$\sin^2\theta + 3\cos^2\theta = 5\sin\theta$ సమీకరణం యొక్క సాధారణ సాధన

Options :

$$n\pi \pm \frac{\pi}{3}, n \in \mathbb{Z}$$

1. ✖

$$n\pi + (-1)^n \frac{\pi}{6}, n \in \mathbb{Z}$$

2. ✔

$$n\pi \pm \frac{\pi}{6}, n \in \mathbb{Z}$$

3. ✖

$$n\pi + (-1)^n \frac{\pi}{3}, n \in \mathbb{Z}$$

4. ✖

Maths - Question 24

Original paper page 20; original question and option regions

Time : 0



If $\cos^{-1} 2x + \cos^{-1} 3x = \frac{\pi}{3}$ and $4x^2 = \frac{a}{b}$ then $a + b =$

$\cos^{-1} 2x + \cos^{-1} 3x = \frac{\pi}{3}$ మరియు $4x^2 = \frac{a}{b}$ అయితే $a + b =$

Options :

1. ✖ 12

2. ✖ 11

3. ✖ 31

4. ✔ 10

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Maths - Question 25

Original paper page 20; original question and option regions

If $\theta = \sec^{-1}(\cosh u)$, then $u =$

$\theta = \sec^{-1}(\cosh u)$, $\Rightarrow \cosh u = \sec \theta$

Options :

$$\log_e \left(\cot \left(\frac{\theta}{2} - \frac{\pi}{4} \right) \right)$$

1. ✖

$$\log_e \left(\tan \left(\frac{\theta}{2} - \frac{\pi}{4} \right) \right)$$

2. ✖

Maths - Question 25 (continued)

Original paper page 21; original question and option regions



$$\log_e \left(\tan \left(\frac{\pi}{4} - \frac{\theta}{2} \right) \right)$$

$$\log_e \left(\tan \left(\frac{\pi}{4} + \frac{\theta}{2} \right) \right)$$

Maths - Question 26

Original paper page 21; original question and option regions

In ΔABC , if $4r_1 = 5r_2 = 6r_3$, then $\sin^2 \frac{A}{2} + \sin^2 \frac{B}{2} + \sin^2 \frac{C}{2} =$

ΔABC లో, $4r_1 = 5r_2 = 6r_3$ అయితే $\sin^2 \frac{A}{2} + \sin^2 \frac{B}{2} + \sin^2 \frac{C}{2} =$

Options :

1. ✖

$$\frac{19}{22}$$

2. ✔

$$\frac{25}{33}$$

3. ✖

$$\frac{74}{99}$$

4. ✖

$$\frac{28}{33}$$

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Maths - Question 26 (continued)

Original paper page 22; original question and option regions



Maths - Question 27

Original paper page 22; original question and option regions

$$\text{In } \Delta ABC, rr_1 \cot \frac{A}{2} + rr_2 \cot \frac{B}{2} + rr_3 \cot \frac{C}{2} =$$

$$\Delta ABC \text{ S } rr_1 \cot \frac{A}{2} + rr_2 \cot \frac{B}{2} + rr_3 \cot \frac{C}{2} =$$

Options :

1. ✓ 3Δ

2. ✗ $3S$

3. ✗ $\frac{S}{\Delta}$

4. ✗ Δ

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Maths - Question 28

Original paper page 22; original question and option regions

$$\text{In } \triangle ABC, bc - r_2 r_3 =$$

$$\triangle ABC \text{ } \mathcal{S}^{\delta} bc - r_2 r_3 =$$

Options :

1.  rr_1

Maths - Question 28 (continued)

Original paper page 23; original question and option regions



$$rr_2$$

$$r_1$$

$$ar_1$$

Maths - Question 29

Original paper page 23; original question and option regions

The angle between the diagonals of the parallelogram whose adjacent sides are

$$2i + 4j - 5k, i + 2j + 3k \text{ is}$$

$2i + 4j - 5k, i + 2j + 3k$ సదిశలు ఆసన్నభుజాలుగా గల సమాంతర చతుర్భుజం యొక్క వికర్ణాల మధ్య గల కోణం

Options :

$$\cos^{-1}\left(\frac{7}{\sqrt{69}}\right)$$

1. ✖

$$\cos^{-1}\left(\frac{1}{7\sqrt{69}}\right)$$

2. ✖

$$\cos^{-1}\left(\frac{1}{7}\right)$$

3. ✖

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Maths - Question 29 (continued)

Original paper page 24; original question and option regions

$$\cos^{-1}\left(\frac{31}{7\sqrt{69}}\right)$$



Maths - Question 30

Original paper page 24; original question and option regions

If the points having the position vectors $-i + 4j - 4k$, $3i + 2j - 5k$, $-3i + 8j - 5k$ and $-3i + 2j + \lambda k$ are coplanar, then $\lambda =$

$-i + 4j - 4k$, $3i + 2j - 5k$, $-3i + 8j - 5k$ మరియు $-3i + 2j + \lambda k$ స్థాన సదిశలుగా గల బిందువులు సతలీయాలైతే, $\lambda =$

Options :

1. ✖ 1

2. ✖ 2

3. ✔ -2

4. ✖ -3

Maths - Question 32

Original paper page 25; original question and option regions

If $\vec{a}, \vec{b}, \vec{c}$ are three vectors such that $|\vec{a}| = |\vec{b}| = |\vec{c}| = \sqrt{3}$ and

$$(\vec{a} + \vec{b} - \vec{c})^2 + (\vec{b} + \vec{c} - \vec{a})^2 + (\vec{c} + \vec{a} - \vec{b})^2 = 36, \text{ then } |2\vec{a} - 3\vec{b} + 2\vec{c}|^2 =$$

$\vec{a}, \vec{b}, \vec{c}$ అనే మూడు సదిశలు $|\vec{a}| = |\vec{b}| = |\vec{c}| = \sqrt{3}$ మరియు

$$(\vec{a} + \vec{b} - \vec{c})^2 + (\vec{b} + \vec{c} - \vec{a})^2 + (\vec{c} + \vec{a} - \vec{b})^2 = 36 \text{ అయ్యేటట్లు ఉంటే, } |2\vec{a} - 3\vec{b} + 2\vec{c}|^2 =$$

Options :

1. ✖ 15

Maths - Question 32 (continued)

Original paper page 26; original question and option regions

25



3. ✖

147

4. ✔

75

Maths - Question 33

Original paper page 26; original question and option regions

The angle between the line with the direction ratios $(2, 5, 1)$ and the plane $8x + 2y - z = 4$ is

$(2, 5, 1)$ దిక్ నిష్పత్తులుగా గల రేఖకు మరియు $8x + 2y - z = 4$ తలానికి మధ్య కోణం

Options :

1. ✖

$$\cos^{-1}\left(\frac{64}{\sqrt{9804}}\right)$$

2. ✖

$$\sin^{-1}\left(\frac{64}{\sqrt{9804}}\right)$$

3. ✔

$$\sin^{-1}\left(\frac{25}{\sqrt{2070}}\right)$$

4. ✖

$$\cos^{-1}\left(\frac{25}{\sqrt{2070}}\right)$$

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Maths - Question 33 (continued)

Original paper page 27; original question and option regions



Maths - Question 34

Original paper page 27; original question and option regions

If the mean deviation about the mean is m and variance is σ^2 for the following data, then

$$m + \sigma^2 =$$

x	1	3	5	7	9
f	4	24	28	16	8

ఈ క్రింది దత్తాంశమునకు అంక మధ్యమము దృష్ట్యా మధ్యమ విచలనం m మరియు విస్తృతి

$$\sigma^2$$
 అయితే, $m + \sigma^2 =$

x	1	3	5	7	9
f	4	24	28	16	8

Options :

1. ✖ 8

2. ✖ 7.2

3. ✖ $\frac{28}{5}$

4. ✔ 6

Maths - Question 39

Original paper page 32; original question and option regions



For a binomial variate $X \sim B(n, p)$ the difference between the mean and variance is 1 and the difference between their squares is 11. If the probability of $P(x = 2) = m \left(\frac{5}{6}\right)^n$ and $n = 36$ then $m:n =$

ఒక ద్విపద చలరాశి $X \sim B(n, p)$ కి అంకమధ్యమము మరియు విస్తృతుల మధ్య భేదము 1 మరియు వాటి వర్గాల మధ్య భేదము 11. $P(x = 2) = m \left(\frac{5}{6}\right)^n$ మరియు $n = 36$ అయితే, $m:n =$

Options :

1. ✖ 6:5
2. ✔ 7:10
3. ✖ 36:1
4. ✖ 42:25

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Maths - Question 40

Original paper page 32; original question and option regions

The probability that a man failing to hit a target is $\frac{1}{3}$. If he fires 4 times, then the probability that he hits the target at least thrice is

ఒక వ్యక్తి ఒక లక్ష్యాన్ని చేధించలేకపోయే సంభావ్యత $\frac{1}{3}$. అతడు 4 సార్లు కాల్పులు జరిపితే, కనీసం మూడుసార్లు లక్ష్యాన్ని చేధించే సంభావ్యత

Maths - Question 40 (continued)

Original paper page 33; original question and option regions

$$\frac{16}{27}$$

$$\frac{11}{27}$$

$$\frac{8}{81}$$

$$\frac{32}{81}$$

If the origin is shifted to remove the first degree terms from the equation

$2x^2 - 3y^2 + 4xy + 4x + 4y - 14 = 0$ then, with respect to this new co-ordinate system, the transformed equation of $x^2 + y^2 - 3xy + 4y + 3 = 0$ is

$2x^2 - 3y^2 + 4xy + 4x + 4y - 14 = 0$ సమీకరణము నుండి మొదటి తరగతి పదాలను తొలగించేయడానికి మూలబిందువును మార్చగా వచ్చిన నూతన నిరూపక వ్యవస్థ దృష్ట్యా $x^2 + y^2 - 3xy + 4y + 3 = 0$ యొక్క రూపాంతరం చెందిన సమీకరణం

Options :

$$x^2 + y^2 - 3xy - 2x + y + 6 = 0$$

1. ✖

$$x^2 + y^2 - 3xy - 2x + 7y + 3 = 0$$

2. ✖

$$x^2 + y^2 - 3xy - 2x + y + 4 = 0$$

3. ✖

$$x^2 + y^2 - 3xy - 2x + 7y + 4 = 0$$

4. ✔

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Maths - Question 42 (continued)

Original paper page 35; original question and option regions



Maths - Question 43

Original paper page 35; original question and option regions

The circumcentre of the triangle formed by the lines $x + y + 2 = 0$, $2x + y + 8 = 0$ and

$x - y - 2 = 0$ is

$x + y + 2 = 0$, $2x + y + 8 = 0$ మరియు $x - y - 2 = 0$ రేఖలతో ఏర్పడే త్రిభుజము యొక్క పరికేంద్రము

Options :

1. ✖ $(-5, 1)$

2. ✔ $(-4, 0)$

3. ✖ $(0, -2)$

4. ✖ $\left(\frac{-8}{3}, \frac{-2}{3}\right)$

Maths - Question 44

Original paper page 36; original question and option regions



If the line $2x - 3y + 5 = 0$ is the perpendicular bisector of the line segment joining $(1, -2)$ and (α, β) , then $\alpha + \beta =$

$(1, -2)$ మరియు (α, β) లను కలుపు రేఖా ఖండాన్ని సమద్విఖండన చేయు రేఖ

$2x - 3y + 5 = 0$ అయితే $\alpha + \beta =$

Options :

1. ✖ 7

2. ✔ 1

3. ✖ -1

4. ✖ -7

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Maths - Question 45

Original paper page 36; original question and option regions

If the area of the triangle formed by the straight lines $-15x^2 + 4xy + 4y^2 = 0$ and $x = \alpha$ is

200 sq. unit, then $|\alpha| =$

$-15x^2 + 4xy + 4y^2 = 0$ మరియు $x = \alpha$ రేఖలతో ఏర్పడే త్రిభుజ వైశాల్యము 200 చదరపు

యూనిట్లయితే $|\alpha| =$

Options :

10

1. ✓

Maths - Question 45 (continued)

Original paper page 37; original question and option regions

20



3. ✖

$5\sqrt{2}$

4. ✖

40

Maths - Question 46

Original paper page 37; original question and option regions

The equation of straight line passing through the point of intersection of the lines represented by

$$x^2 + 4xy + 3y^2 - 4x - 10y + 3 = 0 \text{ and the point } (2, 2) \text{ is}$$

$x^2 + 4xy + 3y^2 - 4x - 10y + 3 = 0$ సూచించే సరళరేఖల ఖండన బిందువు మరియు

$(2, 2)$ బిందువు గుండా పోయేరేఖా సమీకరణము

Options :

1. ✖

$$2x + 3y - 10 = 0$$

2. ✔

$$3x + 2y - 10 = 0$$

3. ✖

$$2x + y - 6 = 0$$

4. ✖

$$x + 2y - 6 = 0$$

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Maths - Question 46 (continued)

Original paper page 38; original question and option regions



Maths - Question 47

Original paper page 38; original question and option regions

The largest among the distances from the point $P(15,9)$ to the points on the circle $x^2 + y^2 - 6x - 8y - 11 = 0$ is

$P(15,9)$ నుండి వృత్తము $x^2 + y^2 - 6x - 8y - 11 = 0$ పై గల బిందువులకు గల దూరములలో గరిష్ఠ దూరం

Options :

1. ✖ 12

2. ✖ 13

3. ✔ 19

4. ✖ 7

Maths - Question 48

Original paper page 38; original question and option regions

The circle $x^2 + y^2 - 8x - 12y + \alpha = 0$ lies in the first quadrant without touching the coordinate axes. If $(6, 6)$ is an interior point to the circle, then

వృత్తము $x^2 + y^2 - 8x - 12y + \alpha = 0$ నిరూపక అక్షాలను తాకకుండా మొదటి పాదంలో ఉంది. $(6, 6)$ బిందువు ఆ వృత్తం యొక్క ఒక అంతర బిందువు అయితే, అప్పుడు

Maths - Question 48 (continued)

Original paper page 39; original question and option regions

$$4 < \alpha < 6$$

$$6 < \alpha < 16$$

$$16 < \alpha < 48$$

$$36 < \alpha < 48$$

Maths - Question 49

Original paper page 39; original question and option regions

The equation of the circle whose diameter is the common chord of the circles

$$x^2 + y^2 - 6x - 7 = 0 \text{ and } x^2 + y^2 - 10x + 16 = 0, \text{ is}$$

$x^2 + y^2 - 6x - 7 = 0$ మరియు $x^2 + y^2 - 10x + 16 = 0$ వృత్తాల ఉమ్మడి జ్యా ను వ్యాసముగా గలిగిన వృత్తము యొక్క సమీకరణము

Options :

$$8x^2 + 8y^2 - 92x + 197 = 0$$

1. ✓

$$x^2 + y^2 - 23x + 197 = 0$$

2. ✗

$$x^2 + y^2 - \frac{23}{2}x + \frac{197}{4} = 0$$

3. ✗

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Maths - Question 49 (continued)

Original paper page 40; original question and option regions



$$4x^2 + 4y^2 - 46x + 197 = 0$$

If the locus of the mid points of the chords of the circle $x^2 + y^2 = 25$, which subtend a right angle at the origin is given by $\frac{x^2}{\alpha^2} + \frac{y^2}{\alpha^2} = 1$, then $|\alpha| =$

$x^2 + y^2 = 25$ అనే వృత్తానికి గీసిన మూల బిందువు దగ్గర లంబకోణాన్ని చేసే జ్యా ల యొక్క మధ్య బిందువుల బిందుపథం $\frac{x^2}{\alpha^2} + \frac{y^2}{\alpha^2} = 1$ అయితే $|\alpha| =$

Options :

1. ✖ $\frac{2}{5}$

2. ✔ $\frac{5}{\sqrt{2}}$

3. ✖ $\frac{2}{25}$

4. ✖ $5\sqrt{2}$

Mandatory : No Calculator : None Response Time : N.A Think Time : N.A 
Time : 0

The radical centre of the circles $x^2 + y^2 + 2x + 3y + 1 = 0$, $x^2 + y^2 + x - y + 3 = 0$,

$x^2 + y^2 - 3x + 2y + 5 = 0$ is

$x^2 + y^2 + 2x + 3y + 1 = 0$, $x^2 + y^2 + x - y + 3 = 0$, $x^2 + y^2 - 3x + 2y + 5 = 0$ అనే

వృత్తాల యొక్క మూల కేంద్రము

Options :

1. ✖ $\left(-\frac{7}{38}, \frac{6}{19}\right)$

2. ✖ $\left(\frac{6}{19}, \frac{14}{19}\right)$

3. ✔ $\left(\frac{14}{19}, \frac{6}{19}\right)$

4. ✖ $\left(\frac{2}{19}, \frac{3}{19}\right)$

Maths - Question 52

Original paper page 41; original question and option regions

Equation of a tangent line of the parabola $y^2 = 8x$, which passes through the point $(1, 3)$ is

$(1, 3)$ గుండా పోయే $y^2 = 8x$ పరావలయం యొక్క ఒక స్పర్శ రేఖ సమీకరణం

Maths - Question 52 (continued)

Original paper page 42; original question and option regions



$$y = 2x + 1$$



$$2y = x + 5$$



$$y = -2x + 5$$



$$2y = 3x + 3$$



Maths - Question 53

Original paper page 42; original question and option regions

If the chord of the ellipse $\frac{x^2}{4} + \frac{y^2}{9} = 1$ having $(1, 1)$ as its middle point is $x + \alpha y = \beta$, then

$x + \alpha y = \beta$ అనేది $\frac{x^2}{4} + \frac{y^2}{9} = 1$ దీర్ఘవృత్తానికి $(1, 1)$ మధ్య బిందువుగా గల జ్యా అయితే, అప్పుడు

Options :

$$\alpha + \beta = 1$$

1. ✖

$$\alpha + 1 = \beta$$

2. ✔

$$\alpha - 1 = \beta$$

3. ✖

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Maths - Question 53 (continued)

Original paper page 43; original question and option regions

$$2\alpha - 1 = 3\beta$$



Maths - Question 54

Original paper page 43; original question and option regions

If a directrix of a hyperbola centred at the origin and passing through the point $(4, -2\sqrt{3})$ is

$\sqrt{5}x = 4$ and e is its eccentricity, then $e^2 =$

మూలబిందువు వద్ద కేంద్రాన్ని కలిగి మరియు $(4, -2\sqrt{3})$ బిందువు గుండా పోయే

అతిపరావలయం యొక్క ఒక నియతరేఖ $\sqrt{5}x = 4$ మరియు దాని ఉత్కేంద్రత e అయితే,

$e^2 =$

Options :

1. ✖

$$\frac{\sqrt{7}}{2}$$

2. ✔

$$\frac{7}{2}$$

3. ✖

$$\frac{35}{4}$$

4. ✖

$$2\sqrt{3}$$

Time : 0

If l_1 and l_2 are the lengths of the perpendiculars drawn from a point on the hyperbola

$$5x^2 - 4y^2 - 20 = 0 \text{ to its asymptotes, then } \frac{l_1^2 l_2^2}{100} =$$

దీర్ఘవృత్తం $5x^2 - 4y^2 - 20 = 0$ పై గల ఒక బిందువు నుండి దాని అనంతస్పర్శరేఖ లకు

గీచిన లంబరేఖల పొడవులు l_1 మరియు l_2 అయితే, $\frac{l_1^2 l_2^2}{100} =$

Options :

1. ✖ $\frac{20}{9}$

2. ✖ $\frac{16}{81}$

3. ✔ $\frac{4}{81}$

4. ✖ $\frac{2}{9}$

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Maths - Question 56

Original paper page 44; original question and option regions

If $O (0, 0, 0)$, $A (3, 0, 0)$, $B (0, 4, 0)$ form a triangle then the incentre of triangle OAB is

$O (0, 0, 0)$, $A (3, 0, 0)$, $B (0, 4, 0)$ లతో ఏర్పడే త్రిభుజం యొక్క అంతరకేంద్రం

Maths - Question 56 (continued)

Original paper page 45; original question and option regions

$$(0, 1, 0)$$

$$(0, 1, 1)$$

$$(1, 0, 1)$$

$$(1, 1, 0)$$

Maths - Question 57

Original paper page 45; original question and option regions

The direction cosines of the line of intersection of the planes $x + 2y + z - 4 = 0$ and $2x - y + z - 3 = 0$ are

$x + 2y + z - 4 = 0$ మరియు $2x - y + z - 3 = 0$ తలల ఖండన రేఖ యొక్క దిక్ కోసైన్ లు

Options :

$$\left(\frac{3}{\sqrt{26}}, \frac{1}{\sqrt{26}}, \frac{-4}{\sqrt{26}} \right)$$

1. ✖

$$\left(\frac{3}{\sqrt{14}}, \frac{2}{\sqrt{14}}, \frac{-1}{\sqrt{14}} \right)$$

2. ✖

Maths - Question 57 (continued)

Original paper page 46; original question and option regions

$$\left(\frac{3}{\sqrt{35}}, \frac{1}{\sqrt{35}}, \frac{-5}{\sqrt{35}} \right)$$

$$\left(\frac{3}{\sqrt{22}}, \frac{-2}{\sqrt{22}}, \frac{3}{\sqrt{22}} \right)$$



Maths - Question 58

Original paper page 46; original question and option regions

If L_1 and L_2 are two lines which pass through origin and having direction ratios $(3, 1, -5)$ and $(2, 3, -1)$ respectively, then equation of the plane containing L_1 and L_2 is

మూలబిందువు గుండాపోతూ $(3, 1, -5)$ మరియు $(2, 3, -1)$ లు దిక్ సంఖ్యలుగా గల రేఖలు వరుసగా L_1 మరియు L_2 అయితే, L_1 మరియు L_2 రేఖలను కలిగియున్న తలం సమీకరణము

Options :

$$4x + 5y - 6z = 0$$

1. ✖

$$5x - y + 3z = 0$$

2. ✖

$$2x - y + z = 0$$

3. ✔

$$x - 5y + 3z = 0$$

4. ✖

Question Number : 59 Question Id : 4509386779 Display Question Number : Yes Is Question

Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Inst

Time : 0

$$\lim_{x \rightarrow \frac{\pi}{4}} \frac{4\sqrt{2} - (\cos x + \sin x)^5}{1 - \sin 2x} =$$

Options :

1. ✓ $5\sqrt{2}$

2. ✗ $3\sqrt{2}$

3. ✗ $2\sqrt{2}$

4. ✗ $\sqrt{2}$

Maths - Question 60

Original paper page 47; original question and option regions

$$\text{If } \lim_{x \rightarrow 0} \frac{e^x - a - \log(1+x)}{\sin x} = 0, \text{ then } a =$$

$$\lim_{x \rightarrow 0} \frac{e^x - a - \log(1+x)}{\sin x} = 0 \text{ అయిన } a =$$

Options :

2

1. ✖

Maths - Question 60 (continued)

Original paper page 48; original question and option regions

0

-1

1



Maths - Question 61

Original paper page 48; original question and option regions

The values of a and b for which the function $f(x) = \begin{cases} 1 + |\sin x|^{a/|\sin x|}, & -\frac{\pi}{6} < x < 0 \\ b, & x = 0 \\ e^{\tan 2x / \tan 3x}, & 0 < x < \frac{\pi}{6} \end{cases}$

is continuous at $x = 0$ are

$x = 0$ వద్ద ప్రమేయం $f(x) = \begin{cases} 1 + |\sin x|^{a/|\sin x|}, & -\frac{\pi}{6} < x < 0 \\ b, & x = 0 \\ e^{\tan 2x / \tan 3x}, & 0 < x < \frac{\pi}{6} \end{cases}$ అవిచ్ఛిన్నం అయ్యేటట్లు

ఉండే a మరియు b ల విలువలు

Options :

$$a = 1, b = \frac{2}{3}$$

1. ✖

$$a = \frac{2}{3}, b = e^{2/3}$$

2. ✔

Maths - Question 61 (continued)

Original paper page 49; original question and option regions



$$a = \frac{2}{3}, b = \frac{3}{2}$$

$$a = -1, b = e^{2/3}$$

4. ✖

Maths - Question 62

Original paper page 49; original question and option regions

If $f(x) = \begin{cases} 2x + 3, & x \leq 1 \\ ax^2 + bx, & x > 1 \end{cases}$ is differentiable $\forall x \in \mathbb{R}$, then $f'(2) =$

$f(x) = \begin{cases} 2x + 3, & x \leq 1 \\ ax^2 + bx, & x > 1 \end{cases}$ అన్ని $x \in \mathbb{R}$ లకు అవకలనీయం అయితే, అప్పుడు $f'(2) =$

Options :

1. ✖ 5

2. ✖ 4

3. ✔ -4

4. ✖ -10

Mandatory : No Calculator : None Response Time : N.A Think Time : N.A**Time : 0**

If $y = t^2 + t^3$ and $x = t - t^4$ then $\frac{d^2y}{dx^2}$ at $t = 1$ is

$y = t^2 + t^3$ మరియు $x = t - t^4$ అయితే, అప్పుడు $t = 1$ వద్ద $\frac{d^2y}{dx^2} =$

Options :

1. ✖ $^{-2/3}$

2. ✔ $^{-4/3}$

3. ✖ $^{8/3}$

4. ✖ 4

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Maths - Question 65

Original paper page 51; original question and option regions

p_1 and p_2 are the perpendicular distances from the origin to the tangent and normal drawn at any point on the curve $x^{2/3} + y^{2/3} = a^{2/3}$ respectively. If $k_1 p_1^2 + k_2 p_2^2 = a^2$, then $k_1 + k_2 =$

p_1 మరియు p_2 లు వరుసగా మూలబిందువు నుండి వక్రము $x^{2/3} + y^{2/3} = a^{2/3}$ పై గల ఏదైనా బిందువు వద్ద గీచిన స్పర్శరేఖ మరియు అభిలంబరేఖలకు గల లంబదూరాలు.

$k_1 p_1^2 + k_2 p_2^2 = a^2$ అయితే, అప్పుడు $k_1 + k_2 =$

Options :

1. ✖

7

2. ✖

6

3. ✔

5

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Maths - Question 65 (continued)

Original paper page 52; original question and option regions



The length of the subnormal at any point on the curve $y = \left(\frac{x}{2024}\right)^k$ is constant if the value of k is

$y = \left(\frac{x}{2024}\right)^k$ అనే వక్రం పై ఏ బిందువు వద్ద నైన ఉపలంబఖండం స్థిరము కావడానికి k యొక్క విలువ

Options :

1. ✖ 1

2. ✖ $\frac{1}{3}$

3. ✔ $\frac{1}{2}$

4. ✖ 0

Maths - Question 67

Original paper page 53; original question and option regions



The acute angle between the curves $x^2 + y^2 = x + y$ and $x^2 + y^2 = 2y$ is

వక్రములు $x^2 + y^2 = x + y$ మరియు $x^2 + y^2 = 2y$ ల మధ్య గల అల్పకోణం

Options :

1. ✖ $\frac{2\pi}{3}$

2. ✖ $\frac{\pi}{2}$

3. ✖ $\frac{\pi}{3}$

4. ✔ $\frac{\pi}{4}$

dia.com

Maths - Question 68

Original paper page 53; original question and option regions

A value of C according to the Lagrange's mean value theorem for $f(x) = (x-1)(x-2)(x-3)$ in $[0, 4]$ is

$[0, 4]$ లో $f(x) = (x-1)(x-2)(x-3)$ కి లెగ్రాంజ్ మధ్యమ విలువ సిద్ధాంతం ప్రకారము వచ్చే C యొక్క విలువలలో ఒకటి

Options :

$$2 + \frac{2}{\sqrt{3}}$$

1. ✓

Maths - Question 68 (continued)

Original paper page 54; original question and option regions



$$2 - \sqrt{\frac{16}{3}}$$

$$1 + \sqrt{\frac{5}{4}}$$

$$2 + \sqrt{\frac{8}{3}}$$



Maths - Question 69

Original paper page 54; original question and option regions

$$\int \frac{dx}{x(x^4 + 1)} =$$

Options :

$$\log \left(\frac{x}{x^4 + 1} \right) + c$$

1. ✖

$$\frac{3}{4} \log (x^4 + 1) + c$$

2. ✖

$$\frac{1}{3} \log \left(\frac{x^3}{x^4 + 1} \right) + c$$

3. ✖

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Maths - Question 69 (continued)

Original paper page 55; original question and option regions

$$\frac{1}{4} \log \left(\frac{x^4}{x^4 + 1} \right) + c$$



Maths - Question 70

Original paper page 55; original question and option regions

$$\int \frac{dx}{\sqrt{\sin^3 x \cos(x - \alpha)}} =$$

Options :

$$\frac{1}{\cos \alpha} \sqrt{\cot x + \tan \alpha} + c$$

1. ✖

$$\frac{1}{\sqrt{\cos \alpha}} \sqrt{\cot x - \tan \alpha} + c$$

2. ✖

$$\frac{-1}{\sqrt{\sin \alpha}} \sqrt{\cot x + \tan \alpha} + c$$

3. ✖

$$\frac{-2}{\sqrt{\cos \alpha}} \sqrt{\cot x + \tan \alpha} + c$$

4. ✔

Maths - Question 71

Original paper page 56; original question and option regions



$$\int \frac{e^{2x}}{\sqrt[4]{e^x + 1}} dx =$$

Options :

1. ✖ $\frac{4}{7} (e^x + 1)^{4/3} (3e^x - 1) + c$

2. ✖ $\frac{2}{21} (e^x + 1)^{3/4} (3e^x - 7) + c$

3. ✔ $\frac{4}{21} (e^x + 1)^{3/4} (3e^x - 4) + c$

4. ✖ $\frac{8}{21} (e^x + 1)^{3/4} (3e^x - 1) + c$

india.com

Maths - Question 72

Original paper page 56; original question and option regions

$$\int \frac{2 - \sin x}{2 \cos x + 3} dx =$$

Options:

$$\frac{2}{\sqrt{5}} \tan^{-1} \left(\frac{1}{\sqrt{3}} \tan \frac{x}{2} \right) - \log \sqrt{2 \cos x + 3} + c$$

1. ✖

Maths - Question 72 (continued)

Original paper page 57; original question and option regions



$$\frac{4}{\sqrt{5}} \tan^{-1} \left(\frac{1}{\sqrt{5}} \tan \frac{x}{2} \right) + \log \sqrt{2 \cos x + 3} + c$$

$$\frac{3}{\sqrt{5}} \tan^{-1} \left(\frac{1}{\sqrt{5}} \tan \frac{x}{2} \right) + \log \sqrt{2 \cos x - 3} + c$$

3. ✖

$$\frac{1}{\sqrt{5}} \tan^{-1} \left(\frac{1}{\sqrt{5}} \tan \frac{x}{3} \right) - \log \sqrt{2 \cos x - 3} + c$$

4. ✖

Maths - Question 73

Original paper page 57; original question and option regions

$$\int \sin^{-1} \sqrt{\frac{x}{a+x}} dx =$$

Options :

$$(a+x) \tan^{-1} \sqrt{\frac{x}{a}} - \sqrt{ax} + c$$

1. ✓

$$\frac{1}{a+x} \tan^{-1} \left(\frac{x}{a} \right) - \sqrt{ax} + c$$

2. ✗

$$(a+x) \tan^{-1} \left(\frac{a}{x} \right) + \sqrt{ax} + c$$

3. ✗

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Maths - Question 73 (continued)

Original paper page 58; original question and option regions

$$\sqrt{a+x} \tan^{-1} \left(\frac{x}{a} \right) + ax + c$$



Maths - Question 74

Original paper page 58; original question and option regions

$$\int_{-1/24}^{1/24} \sec x \log \left(\frac{1-x}{1+x} \right) dx =$$

Options :

1. ✖ $\frac{\pi}{2}$

2. ✖ π

3. ✖ 1

4. ✔ 0

Maths - Question 75

Original paper page 58; original question and option regions

If $[x]$ is the greatest integer function, then $\int_0^5 [x] dx =$

$[x]$ అనేది గరిష్ఠ పూర్ణాంక ప్రమేయము అయితే $\int_0^5 [x] dx =$

Maths - Question 75 (continued)

Original paper page 59; original question and option regions

ons :



15

2

3

10

Maths - Question 76

Original paper page 59; original question and option regions

$$\int_0^{\pi/2} \frac{1}{1+\sqrt{\tan x}} dx =$$

Options :

1. ✖ 0

2. ✖ $\frac{\pi}{2}$

3. ✖ $\frac{\pi}{3}$

4. ✔ $\frac{\pi}{4}$

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Maths - Question 76 (continued)

Original paper page 60; original question and option regions



Maths - Question 77

Original paper page 60; original question and option regions

$$\int_0^{\pi} \frac{x \sin x}{1 + \cos^2 x} dx =$$

Options :

1. ✖ 0

2. ✖ $\frac{\pi}{2}$

3. ✖ $\frac{\pi^2}{2}$

4. ✔ $\frac{\pi^2}{4}$

Maths - Question 78

Original paper page 60; original question and option regions

Order and degree of the differential equation $\frac{d^3y}{dx^3} = \left[1 + \left(\frac{dy}{dx}\right)^2\right]^{5/2}$ respectively are

$\frac{d^3y}{dx^3} = \left[1 + \left(\frac{dy}{dx}\right)^2\right]^{5/2}$ అనే అవకలన సమీకరణము యొక్క తరగతి, పరిమాణములు వరుసగా

Maths - Question 78 (continued)

Original paper page 61; original question and option regions



5, 2

2. ✖

3, 5

3. ✔

3, 2

4. ✖

2, 3

The general solution of the differential equation $\left(x \sin \frac{y}{x}\right) dy = \left(y \sin \frac{y}{x} - x\right) dx$ is

$\left(x \sin \frac{y}{x}\right) dy = \left(y \sin \frac{y}{x} - x\right) dx$ అవకలన సమీకరణం యొక్క సాధారణ సాధన

Options :

1. ✖ $\cos \frac{x}{y} = \log_e x + c$

2. ✔ $\cos \frac{y}{x} = \log_e x + c$

3. ✖ $\cos \frac{x}{y} = \log_e y + c$

4. ✖ $\cos \frac{y}{x} = \log_e y + c$

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Maths - Question 80 (continued)

Original paper page 63; original question and option regions



Maths - Question 1

Original paper page 2; original question and option regions

The domain of the real valued function $f(x) = \log_2 \log_3 \log_5 (x^2 - 5x + 11)$ is

$f(x) = \log_2 \log_3 \log_5 (x^2 - 5x + 11)$ అనే వాస్తవ మూల్య ప్రమేయం యొక్క ప్రదేశం

Options :

$(2, \infty)$

1. ✖

$(-\infty, 3)$

2. ✖

Maths - Question 1 (continued)

Original paper page 3; original question and option regions

$$(2, 3)$$

$$(-\infty, 2) \cup (3, \infty)$$



Maths - Question 2

Original paper page 3; original question and option regions

The range of the real valued function $f(x) = \frac{x^2+2x-15}{2x^2+13x+15}$ is

$f(x) = \frac{x^2+2x-15}{2x^2+13x+15}$ అనే వాస్తవ మూల్య ప్రమేయం యొక్క వ్యాప్తి

Options :

$$\mathbb{R} - \left\{-5, -\frac{3}{2}\right\}$$

1. ✖

$$\mathbb{R} - \left\{-5, \frac{1}{2}\right\}$$

2. ✖

$$\mathbb{R} - \left\{\frac{1}{2}, \frac{8}{7}\right\}$$

3. ✔

$$\mathbb{R} - \left\{-\frac{3}{2}, \frac{8}{7}\right\}$$

4. ✖

Question Number : 3 Question Id : 4509386883 Display Question Number : Yes Is Question

Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Inst

Time : 0

$$\frac{1}{1.5} + \frac{1}{5.9} + \frac{1}{9.13} + \dots \text{ upto } n \text{ terms} =$$

$$\frac{1}{1.5} + \frac{1}{5.9} + \frac{1}{9.13} + \dots (n \text{ పదాల వరకు}) =$$

Options :

1. ✖ $\frac{1}{4n+1}$

2. ✖ $\frac{4}{4n+1}$

3. ✔ $\frac{n}{4n+1}$

4. ✖ $\frac{4n+1}{5(4n+1)}$

Maths - Question 4

Original paper page 5; original question and option regions



If $A = \begin{bmatrix} 2 & 3 & 4 \\ 1 & k & 2 \\ 4 & 1 & 5 \end{bmatrix}$ is singular matrix, then the quadratic equation having the roots k and $\frac{1}{k}$ is

$A = \begin{bmatrix} 2 & 3 & 4 \\ 1 & k & 2 \\ 4 & 1 & 5 \end{bmatrix}$ ఒక అసాధారణ మాత్రిక అయితే, k మరియు $\frac{1}{k}$ మూలాలుగా గల వర్గ సమీకరణము

Options :

1. ✖ $6x^2 + 13x + 6 = 0$

2. ✖ $12x^2 - 25x + 12 = 0$

3. ✔ $6x^2 - 13x + 6 = 0$

4. ✖ $2x^2 - 5x + 2 = 0$

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Maths - Question 5

Original paper page 5; original question and option regions

Let A be a 4×4 matrix and P be its adjoint matrix. If $|P| = \left| \frac{A}{2} \right|$, then $|A^{-1}| =$

A అనేది 4×4 మాత్రిక మరియు P అనేది A యొక్క అనుబంధ మాత్రిక అనుకుందాం.

$|P| = \left| \frac{A}{2} \right|$ అయితే $|A^{-1}| =$

Options :

$$\pm \frac{1}{4}$$

1. ✖

Maths - Question 5 (continued)

Original paper page 6; original question and option regions



± 8

± 2

± 4

Maths - Question 6

Original paper page 6; original question and option regions

The system $x + 2y + 3z = 4, 4x + 5y + 3z = 5, 3x + 4y + 3z = \lambda$ is consistent and $3\lambda = n + 100$, then $n =$

$x + 2y + 3z = 4, 4x + 5y + 3z = 5, 3x + 4y + 3z = \lambda$ సమీకరణ వ్యవస్థ సంగతము మరియు $3\lambda = n + 100$, అయితే $n =$

Options :

1. ✖ - 42

2. ✔ - 86

3. ✖ 16

4. ✖ - 24

Question Number : 7 Question Id : 4509386887 Display Question Number : Yes Is Question

Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Inst
Time : 0

The complex conjugate of $(4 - 3i)(2 + 3i)(1 + 4i)$ is

$(4 - 3i)(2 + 3i)(1 + 4i)$ యొక్క సంకీర్ణ సంయుగ్మం

Options :

1. ✖ $7 + 74i$

2. ✖ $-7 + 74i$

3. ✔ $-7 - 74i$

4. ✖ $7 - 74i$

Maths - Question 8

Original paper page 7; original question and option regions

If the amplitude of $(Z - 2)$ is $\frac{\pi}{2}$, then the locus of Z is

$(Z - 2)$ యొక్క ఆయామము $\frac{\pi}{2}$ అయితే, Z యొక్క బిందుపథము

Options :

$x = 0, y > 0$

1. ✖

Maths - Question 8 (continued)

Original paper page 8; original question and option regions

$$x = 2, y > 0$$

$$x > 0, y = 2$$

$$x > 0, y = 0$$



Maths - Question 9

Original paper page 8; original question and option regions

If ω is the cube root of unity, $\frac{a+b\omega+c\omega^2}{c+a\omega+b\omega^2} + \frac{a+b\omega+c\omega^2}{b+c\omega+a\omega^2} =$

ω అనేది 1 యొక్క ఘనమూలం అయిన, $\frac{a+b\omega+c\omega^2}{c+a\omega+b\omega^2} + \frac{a+b\omega+c\omega^2}{b+c\omega+a\omega^2} =$

Options :

1. ✖ 2

2. ✖ -2

3. ✖ 1

4. ✔ -1

Mandatory : No Calculator : None Response Time : N.A Think Time : N.A 
Time : 0

Roots of the equation $a(b - c)x^2 + b(c - a)x + c(a - b) = 0$ are

$a(b - c)x^2 + b(c - a)x + c(a - b) = 0$ యొక్క మూలాలు

Options :

1. ✖ $\frac{a(b - c)}{c(a - b)}, 1$

2. ✖ $\frac{b(c - a)}{c(a - b)}, 1$

3. ✔ $\frac{c(a - b)}{a(b - c)}, 1$

4. ✖ $\frac{c(a - b)}{b(c - a)}, 1$

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Maths - Question 11

Original paper page 9; original question and option regions

If $(3 + i)$ is a root of $x^2 + ax + b = 0$ then $a =$

$x^2 + ax + b = 0$ యొక్క ఒక మూలం $(3 + i)$ అయితే $a =$

Options :

1. ✖ 3

Maths - Question 11 (continued)

Original paper page 10; original question and option regions

—3

✗ 6

✓ —6



Maths - Question 12

Original paper page 10; original question and option regions

The algebraic equation of degree 4 whose roots are the translates of the roots of the equation

$$x^4 + 5x^3 + 6x^2 + 7x + 9 = 0 \text{ by } -1 \text{ is}$$

$x^4 + 5x^3 + 6x^2 + 7x + 9 = 0$ సమీకరణం మూలాలు -1 తో మార్పు చెందగా వచ్చే

విలువలను మూలాలుగా గల 4 వ తరగతి బీజీయ సమీకరణం

Options :

1. ✖ $x^4 + x^3 - 3x^2 + 6x + 4 = 0$

2. ✔ $x^4 + 9x^3 + 27x^2 + 38x + 28 = 0$

3. ✖ $x^4 + 5x^3 + 6x^2 + 7x + 9 = 0$

4. ✖ $x^4 - 5x^3 + 6x^2 - 7x + 9 = 0$

Question Number : 13 Question Id : 4509386893 Display Question Number : Yes Is Question

Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Inst
Time : 0

If the roots of the equation $4x^3 - 12x^2 + 11x + m = 0$ are in arithmetic progression, then $m =$

$4x^3 - 12x^2 + 11x + m = 0$ యొక్క మూలాలు అంకశ్రేణిలో వుంటే $m =$

Options :

1. ✓ -3

2. ✗ 1

3. ✗ 2

4. ✗ 3

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Maths - Question 14

Original paper page 11; original question and option regions

The number of 5 digit odd numbers greater than 40,000 that can be formed by using 3, 4, 5, 6, 7, 0 so that at least one of its digit must be repeated is

3, 4, 5, 6, 7, 0 లను ఉపయోగించి ఏర్పరచగల 40,000 కంటే పెద్దవైన 5 అంకెల బేసి సంఖ్యలలో కనీసం ఒక అంకె అయినా పునరావృతం అయ్యే వాని సంఖ్య

Options :

1. ✖ 2592

2. ✖ 240

Maths - Question 14 (continued)

Original paper page 12; original question and option regions



3032

2352

Maths - Question 15

Original paper page 12; original question and option regions

The number of ways in which 3 men and 3 women can be arranged in a row of 6 seats, such that the first and last seats must be filled by men is

3 పురుషులు మరియు 3 స్త్రీలను, 6 సీట్లు కలిగిన ఒక వరుసలో, మొదట మరియు చివరి సీట్లలో పురుషులు మాత్రమే కూర్చునే విధంగా, అమర్చగలిగే విధాల సంఖ్య

Options :

1. ✖ 720

2. ✖ 36

3. ✔ 144

4. ✖ 72

Maths - Question 16

Original paper page 13; original question and option regions



If a committee of 10 members is to be formed from 8 men and 6 women, then the number of different possible committees in which the men are in majority is

8 మంది పురుషులు మరియు 6 మంది స్త్రీలతో 10 మంది సభ్యులు కలిగిన కమిటీని ఏర్పాటు చేయవలసి వస్తే, పురుషులు అధిక సంఖ్యలో ఉండే విధంగా సాధ్యపడే వివిధ కమిటీల సంఖ్య

Options :

1. ✖ 931

2. ✖ 175

3. ✖ 48

4. ✔ 595

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Maths - Question 17

Original paper page 13; original question and option regions

If the eleventh term in the binomial expansion of $(x + a)^{15}$ is the geometric mean of the eighth and twelfth terms, then the greatest term in the expansion is

$(x + a)^{15}$ యొక్క ద్వీపద విస్తరణలోని 11 వ పదం, 8వ మరియు 12వ పదముల గుణ మధ్యమం అయితే, అప్పుడు ఆ విస్తరణ లో గరిష్ఠ పదము

Options :

7th term

7వ పదము

1. ✖

Maths - Question 17 (continued)

Original paper page 14; original question and option regions

8th term

8వ పదము



9th term

9వ పదము

10th term

10వ పదము

Maths - Question 18

Original paper page 14; original question and option regions

The sum of the rational terms in the binomial expansion of $(\sqrt{2} + 3^{1/5})^{10}$ is

$(\sqrt{2} + 3^{1/5})^{10}$ యొక్క ద్విపద విస్తరణలో అకరణీయ పదముల యొక్క మొత్తము

Options :

1. ✓ 41

2. ✗ 39

3. ✗ 32

4. ✗ 30

Question Number : 19 Question Id : 4509386899 Display Question Number : Yes Is Question

Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Inst

Time : 0

If $\frac{1}{(3x+1)(x-2)} = \frac{A}{3x+1} + \frac{B}{x-2}$ and $\frac{x+1}{(3x+1)(x-2)} = \frac{C}{3x+1} + \frac{D}{x-2}$, then

$\frac{1}{(3x+1)(x-2)} = \frac{A}{3x+1} + \frac{B}{x-2}$ మరియు $\frac{x+1}{(3x+1)(x-2)} = \frac{C}{3x+1} + \frac{D}{x-2}$ అయితే,

Options :

$A + 3B = 0, A:C = 1:3, B:D = 2:3$

1. ✖

$A + 3B = 0, A:C = 3:1, B:D = 3:2$

2. ✖

$A - 3B = 0, A:C = 3:2, B:D = 1:3$

3. ✖

$A + 3B = 0, A:C = 3:2, B:D = 1:3$

4. ✔

Maths - Question 20

Original paper page 15; original question and option regions

If the period of the function $f(x) = \frac{\tan 5x \cos 3x}{\sin 6x}$ is α , then $f\left(\frac{\alpha}{8}\right) =$

$f(x) = \frac{\tan 5x \cos 3x}{\sin 6x}$ యొక్క ఆవర్తనం α అయితే, $f\left(\frac{\alpha}{8}\right) =$

Maths - Question 20 (continued)

Original paper page 16; original question and option regions



1.
 $\frac{1}{2}$

2. ✖ -1

3. ✔ $\frac{1}{\sqrt{2}}$

4. ✖ $-\frac{1}{\sqrt{2}}$

am

Maths - Question 21

Original paper page 16; original question and option regions

If $\sin x + \sin y = \alpha$, $\cos x + \cos y = \beta$ then $\operatorname{cosec}(x+y) =$

$\sin x + \sin y = \alpha$, $\cos x + \cos y = \beta$ అయితే, అప్పుడు $\operatorname{cosec}(x+y) =$

Options :

1. ✖

$$\frac{\beta^2 - \alpha^2}{\beta^2 + \alpha^2}$$

2. ✖

$$\frac{2\alpha\beta}{\beta^2 - \alpha^2}$$

3. ✔

$$\frac{\beta^2 + \alpha^2}{2\alpha\beta}$$

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Maths - Question 21 (continued)

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$$\frac{2\alpha\beta}{\beta^2 + \alpha^2}$$

If $P+Q+R=\frac{\pi}{4}$, then $\cos\left(\frac{\pi}{8}-P\right)+\cos\left(\frac{\pi}{8}-Q\right)+\cos\left(\frac{\pi}{8}-R\right)=$

$P+Q+R=\frac{\pi}{4}$, అయితే, అప్పుడు $\cos\left(\frac{\pi}{8}-P\right)+\cos\left(\frac{\pi}{8}-Q\right)+\cos\left(\frac{\pi}{8}-R\right)=$

Options :

$$4\cos\frac{P}{2}\cos\frac{Q}{2}\cos\frac{R}{2}-\cos\frac{\pi}{8}$$

1. ✓

$$4\cos\frac{P}{2}\cos\frac{Q}{2}\sin\frac{R}{2}+\cos\frac{\pi}{8}$$

2. ✗

$$4\sin\frac{P}{2}\sin\frac{Q}{2}\sin\frac{R}{2}-\cos\frac{\pi}{8}$$

3. ✗

$$4\sin\frac{P}{2}\cos\frac{Q}{2}\sin\frac{R}{2}+\cos\frac{\pi}{8}$$

4. ✗